Passive transport

- Simple diffusion
- Facilitated diffusion (glucose in basolateral membrane)

Active transport

Active reabsorption

- Primary active transport (Sodium-potassium ATPase pump)
- Secondary active transport
- Secondary active reabsorption (glucose by sodium in PT)

Active Secretion

- Primary active Secretion
- Secondary active Secretion (H^+ by sodium in PT)

Osmosis

Pinocytosis

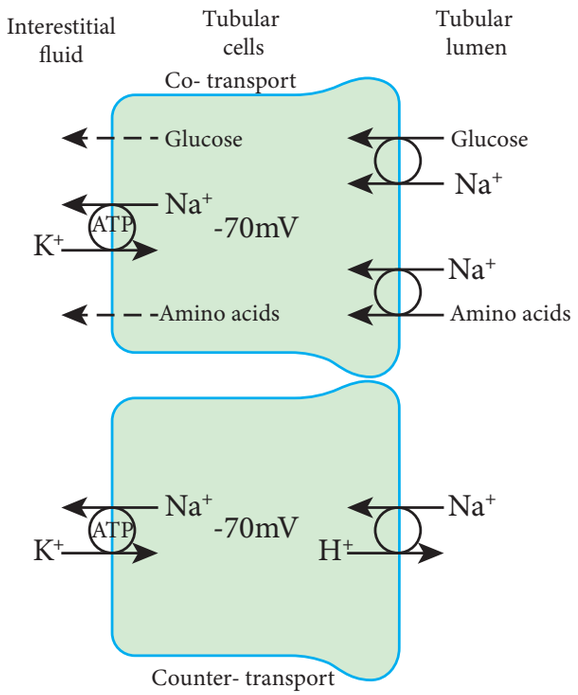


Figure 10: Movement of substances across the renal tubular membrane that aid in urine formation (by filtration, reabsorption,secretion) by kidneys.

Activity

1. Perform the experiment to know the diffusion using coloured liquid or jelly like substance and water that can help you to understand how water, oxygen and small substances are transported across the membrane (Use paper towels as membrane). Do the experiment in the presence and absence of heat i.e hot water and cold water. (Higher temperatures speed up diffusion because molecules have more kinetic energy at higher temperatures).
2. Use potato and sugar solution to understand osmosis. Calculate the difference in height of the solution inside the cut potato (well like pit engraved in potato) which you have kept in a petri dish filled with coloured water.

Summary

1. The cell membrane is an essential feature for the survival and homeostasis of a cell.
2. The membrane regulates the interaction of a cell with its external environment and it allows substances to enter or leave the cell.
3. The membrane is chiefly made up of lipids and proteins; carbohydrates make up a small percentage.

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